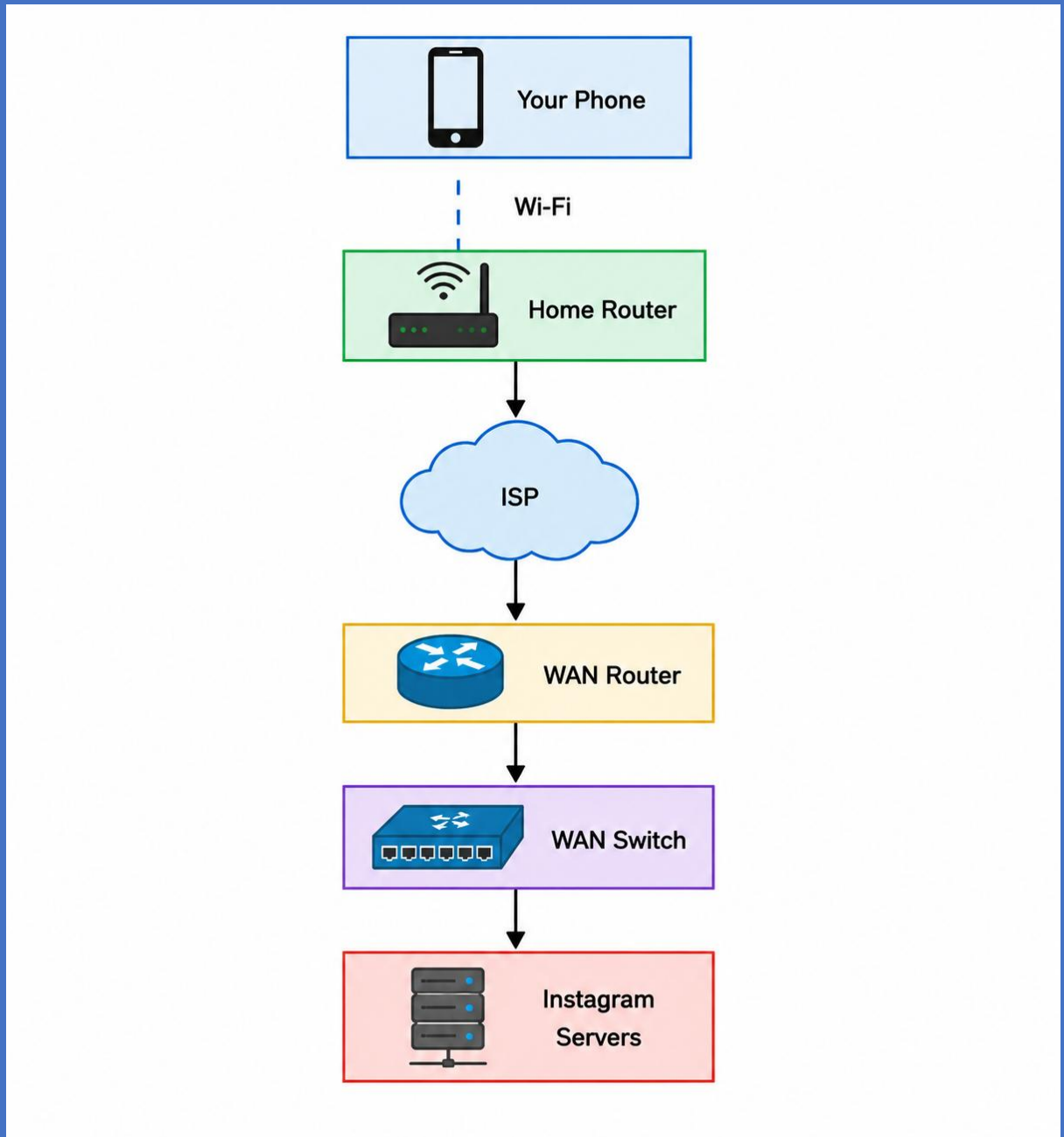


Tasks

1. Draw a simple diagram (on paper or using an online tool like draw.io) showing how your phone connects to Instagram's servers from your home WiFi, including the ISP and at least two Wide Area Network (WAN) devices (like routers or switches) in the path.



2. Create a table listing three types of WAN technologies (such as MPLS, Leased Line, and Broadband Internet) and compare them based on speed, cost, and where you might see them used in real life (e.g., Zomato office, IRCTC ticketing, your home).



WAN Technology	Speed	Cost	Real-World Usage Example
MPLS (Multiprotocol Label Switching)	High (up to several Gbps)	Expensive	Used by Zomato offices for secure, fast data transfer between branches
Leased Line	Very high and stable	Very expensive	Used by IRCTC ticketing systems for constant connectivity and reliability
Broadband Internet	Moderate to high (up to 1 Gbps for fiber)	Affordable	Common in homes for apps like Instagram, YouTube, and online classes

3. Research and write a short explanation of how a VPN (Virtual Private Network) uses WANs to let you access apps like Spotify or YouTube securely from a public WiFi at a coffee shop.
Hint: Focus on how the VPN creates a private tunnel over the public WAN.



How a VPN Uses WAN to Provide Secure Access

A Virtual Private Network (VPN) creates a secure and encrypted connection between a user's device and a VPN server over the Wide Area Network (WAN), which is the Internet. When connected to a public WiFi, such as in a coffee shop, all internet traffic first passes through the encrypted VPN tunnel before reaching websites or apps like Spotify or YouTube.

This encrypted tunnel protects sensitive information from hackers or other users on the same public WiFi network. The VPN server then forwards the request to the destination website and sends the response back through the same secure tunnel.

Using a VPN on public WiFi improves privacy, protects personal data, and allows users to access online services more securely over the WAN.

4. Imagine Flipkart wants to connect its warehouses in Ahmedabad, Mumbai, and Delhi. Suggest which WAN technology would be best for them and justify your answer with two reasons.



Suggested WAN Technology: MPLS (Multiprotocol Label Switching)

Justification:

1. **Reliable and Fast Communication:** MPLS provides a stable, high-speed connection between warehouses, ensuring quick data transfer and smooth business operations.
2. **Secure Private Network:** MPLS uses a private network instead of the public Internet, making it more secure for transferring important business and customer data.